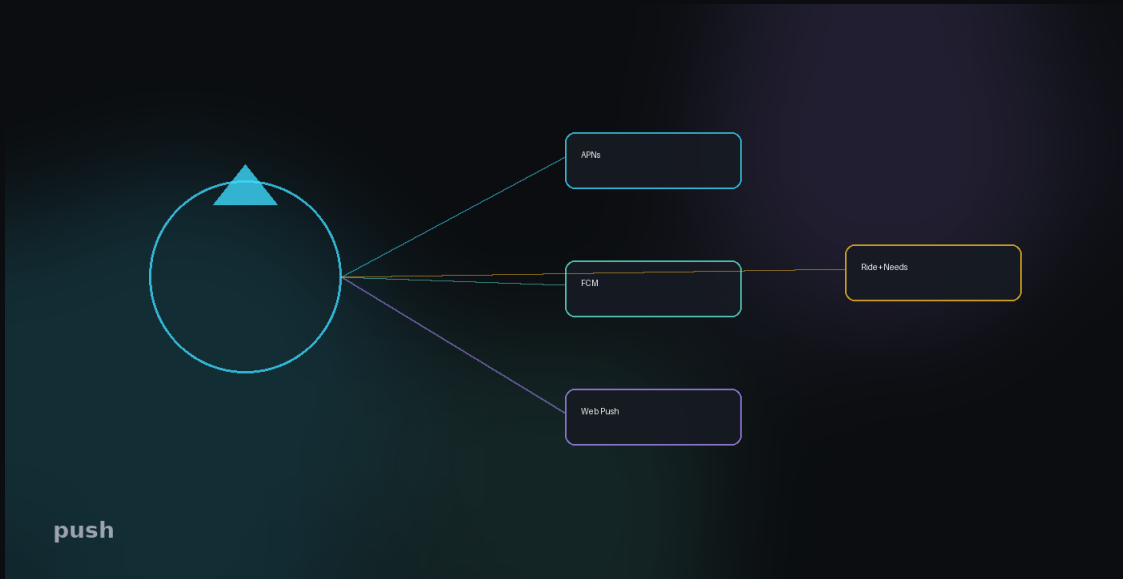




Push Notifications — Research + Roadmap Digest

Dark visual edition · 2026-09-24

Card t_47c87ab8 · NOTIFY · Research digest (no app implementation)

Chuck Broaddus / Kelyra · Confidential

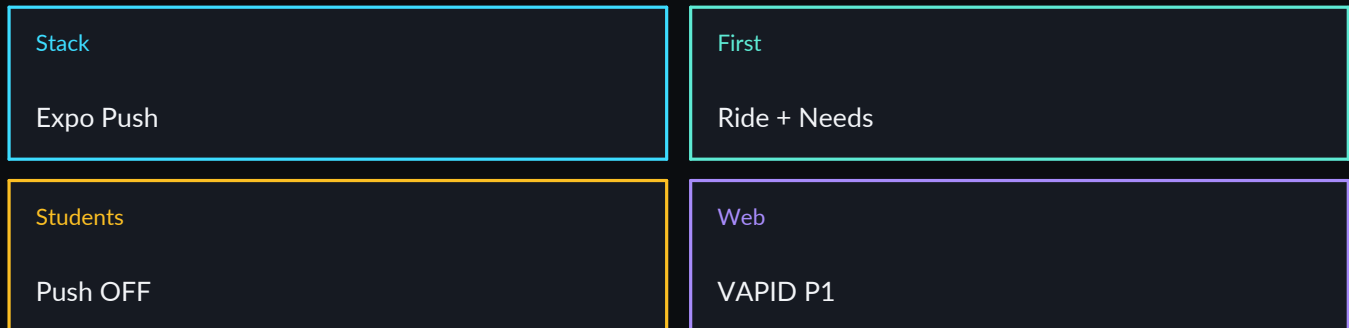
Source: dated research digests under notes/company · graphics illustrate locked recommendations.

Executive recommendation

Locked recommendation

Expo Notifications with APNs + FCM via Expo Push Service; Edge Function send path; Ride + Needs draft-ready first; preference center + quiet hours; payload minimization; student push off. Web Push is P1.

Decision cards



Server-mediated send path

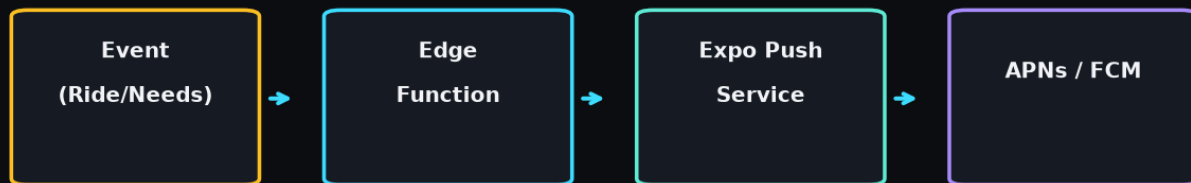


Figure: Event → Edge → Expo → APNs/FCM.

Push Notifications — Research + Roadmap Digest (NOTIFY-R1 / P1)

Date: 2026-09-24 Author: Chief of Staff / Grok Bot (Kelyra) Status: Ready for Chuck review SoT pair: [notes/company/push-notifications-research.md](#) · [notes/company/push-notifications-plan.md](#) Card: [t_47c87ab8](#) Access date for citations: 2026-09-24 (America/Chicago)

This dated file is a readable pack for Chuck. Full detail lives in the SoT pair above. Research + roadmap only — no app implementation. No Hermes staffing.

Top recommendations (P0–P2)

P0 (phones — reverse soft-skip carefully)

- 1. P0-1/2 — Add `expo-notifications` + EAS APNs + FCM v1 credentials; permission UX after a value moment (not splash); Android channels for Ride / Needs / Messages.
- 2. P0-3/4 — Store per-device Expo tokens (RLS); send only from a Supabase Edge Function → Expo Push Service with `EXPO_ACCESS_TOKEN` enhanced security; prune `DeviceNotRegistered`.
- 3. P0-5/8 — Preference center + quiet hours; payload minimization (no scores / IEP / twin mix).
- 4. P0-6/7 — Wire Ride + Needs draft ready first; deep-link into existing surfaces.
- 5. P0-9 — Student push off; no student enable UI.

P1

- Web Push (VAPID) for Chromebook / desk browsers; optional message + calendar reminder pushes; richer mute; ops metrics; office Ride alert alignment.

P2

- Direct APNs/FCM escape hatch; opt-in school topics (never grade topics); policy-gated student push; multi-school credential isolation.

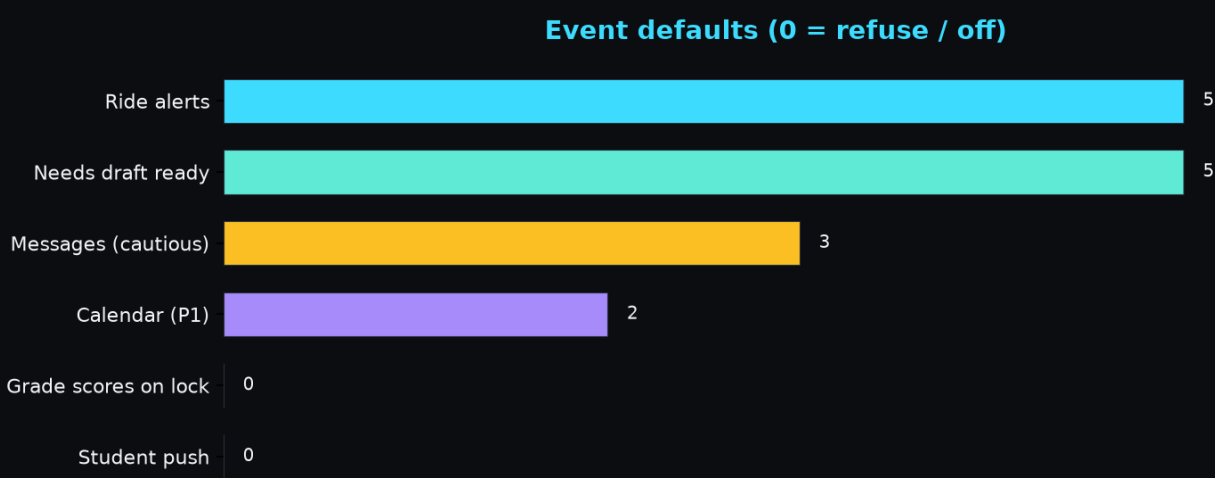


Figure: Event default priorities.

Research (digest)

Verdict

Ship push as a server-mediated alert layer on top of the existing in-app bell — not a new chat product. Best fit for Expo 57 + Supabase: Expo Push Service fan-out to APNs + FCM, with Web Push later for desktop. Architecture today still says Push | Skip in v1 ([docs/architecture.md](#)); this roadmap proposes a controlled reverse for Ride/Needs.

Platforms (one table)

Platform	Mechanism	Kelyra path
iOS / iPadOS	APNs HTTP/2	Via Expo (P0) or direct (P2)
Android	FCM HTTP v1	Via Expo (P0) or direct (P2)
Chrome / Edge / Firefox desktop	Web Push → FCM / Mozilla	P1
Safari macOS 16+	Web Push → APNs	P1
iOS Safari tab	No push	Installed PWA only (limitation)
Native Mac/Win apps	—	Out of scope

Supabase send path

Authorized event → Edge Function → Expo Push API → APNs/FCM

Documented by Supabase (Expo + FCM Edge examples). Kelyra should use a device token table (not a single profile column) and resolve audience with existing hat / parent_students / twin laws before send.

Concrete Kelyra gaps today

- No expo-notifications dependency / plugin.
- No push Edge function.
- No device token / prefs schema.

Privacy locks for lock-screen alerts



Figure: Lock-screen privacy locks.

- In-app /notifications is pull-only.
- EAS project exists (eas.json) but push credentials not part of this research’s live verification (Eng confirms at implement time).

Privacy

Lock screens are bystander-visible → FERPA soft posture: minimize copy; deep-link into RLS screens; COPPA → students off; twins never mixed; OS permission ≠ product consent.

Plan (digest)

Locked decision: Expo Push Service P0; Web Push P1; direct vendor APIs P2.

Event defaults: Ride + Needs on (after opt-in); messages cautious; calendar P1; grades never as lock-screen scores; students off.

Effort guess: P0 ~1–2 Eng weeks post-credentials + 2-device QA; P1 Web Push ~1–2 weeks.

Open questions for Chuck

- 1. Reverse architecture soft-skip for Ride/Needs now?
- 2. Messages in P0 or P1?
- 3. Ride quiet-hours bypass default?
- 4. Web Push required for “other computers” in first paid tier?
- 5. Student forever-off vs later policy?



Figure: P0 / P1 / P2 roadmap.

- 6. Fold `t_c20b5e0d` office banned-parent alert into this epic?

Citations (primary — access 2026-09-24 CT)

- <https://docs.expo.dev/push-notifications/push-notifications-setup/>
- <https://docs.expo.dev/push-notifications/sending-notifications/>
- <https://docs.expo.dev/push-notifications/sending-notifications-custom>
- <https://developer.apple.com/documentation/usernotifications/sending-notification-requests-to-apns>
- <https://firebase.google.com/docs/cloud-messaging/send/v1-api>
- <https://supabase.com/docs/guides/functions/examples/push-notifications>
- <https://developer.mozilla.org/en-US/docs/Web/API/PushSubscription>
- <https://www.w3.org/TR/push-api/>

Full citation list in the research SoT.

Files

Path	Role
notes/company/push-notifications-research.md	Research SoT

Path	Role
notes/company/push-notifications-plan.md	Roadmap / plan SoT
notes/research/2026-09-24-push-notifications-report.md	This digest

Meta · Card t_47c87ab8 (NOTIFY) · Dark visual edition 2026-09-24 · Not legal advice where privacy/compliance topics appear · No Hermes staffing.